

Barnaby O'Brien

07513308584 | barnaby.obrien@gmail.com | linkedin.com/in/barnaby-obrien | github.com/BarnabyOBrien | Oxford, UK

EDUCATION

University of Oxford: MMathPhys [graduating 2026]

Oct. 2022 – Present

- Achieved **1st class** classification (76%)
- Relevant Modules Taken: Machine Learning Fundamentals with Applications to Physics and Mathematics, Linear Algebra (Achieved 92%)
- Awarded a **Scholarship** recognising 'excellent performance in University examinations' by St Hilda's College.

Exeter School: A-Levels and GCSEs

Sep. 2017 – Jun. 2022

- A-Levels: **four A*s** (Maths, Further Maths, Physics, Chemistry).
- **Best in school year** Maths and Physics, awarded **Exhibition** and **Music** Scholarship, Exeter School

RELEVANT EXPERIENCE

Intern, University Of Oxford Mathematics Department

Jun. 2026 – Present

- Researching higher Martin polynomials in combinatorial quantum field theory, with Dr Paul Balduf. Developing code to calculate the combinatorics of Feynman diagrams, using our tensor field theory interpretation

Programming: Data Science Competition

Jul. 2026

- Ranked in the top 3% in the Kaggle 'House Predictions' competition, using **Python**. Implemented a pipeline using **transformer based architecture** which outperformed more traditional machine learning models

Programming: Genetic Algorithm

Jun. 2026

- Developed a genetic algorithm to find allowed potentials for an inflation model, using **Python**. Resulted in selecting a dataset of allowed inflationary potentials from a 10 billion parameter state space

Dissertation: Simulating Cosmic Strings

Aug. 2025 – Jun. 2026

- Simulated cosmic string networks, using **MATLAB**. Demonstrated the novel conclusion that the observation of a kination peak by gravitational wave detectors is unlikely in the near future
- Simulated topological defects in a global U(1) field theory, using **MATLAB**. Redemonstrated their inter-commuting properties, a fundamental result in cosmic string research

Public speaking: Symbolic Regression Neural Networks

Feb. 2024

- Presented a talk on the theory, structure, and applications of symbolic regression neural networks, to an audience of thirty. Demonstrated how symbolic regression can reduce the 'black-box' nature of neural networks

Programming: Building and testing a Neural Network

Jul. 2023 – Sep. 2023

- Programmed (from scratch) an implementation of a neural network to recognize handwritten digits from the MNIST database, using the Python library **NumPy**

ADDITIONAL EXPERIENCE

Intern, Love Productions Ltd.

Sep. 2025

- Worked in both writing and casting teams, developing client focused pitches and facilitating talent recruitment

Teacher, West Exe School Devon

Jun. 2025 - Jul. 2025

- Taught class of thirty at West Exe School in Devon as part of the SWIFT and Blackdown Education internship

Oxford Imps, Improv Comedy Group

Oct. 2024 – Present

- Perform weekly with the Oxford Imps, an international touring improv comedy group, to audiences of 50–100.

Writer, Director and Producer of own original musical

Jul. 2024 – Oct. 2024

- Produced, wrote and directed my own musical 'First Aid', with runs at Burton Taylor Studio Oxford and Edinburgh Fringe 2025, leading a team of six

Intern, Lucky Egg Ltd.

Mar. 2023

- Worked in the development team, creating new potential board game products, analyzing their marketability and target costing

SKILLS

- **Programming:** Proficient in **Python** (NumPy, Pandas, Scikit-Learn, PyTorch), **MATLAB**, **Mathematica**; familiar with **C#**.